

PSF-Zero

An analytic gate synthesizer and a verified, ordering-aware layout search for quantum circuit transpilation

Technical overview — TN Holdings

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Qiskit's own layout search fails silently

Mapping a quantum circuit onto real hardware requires assigning logical qubits to physical ones so every interacting pair is physically connected. Qiskit's VF2Layout pass searches for this assignment before falling back to slower, heuristic routing.

On fully-saturated circuits — every physical qubit in use, exactly the regime that matters for maximizing device utilization — this search has a sharp, reproducible failure region.

It reports “no solution exists” — on instances where a valid layout provably does.

MEASURED FAILURE RATE

19 / 26

saturated test instances fail outright

271X

slower before
failing

9.2s

median time
to that wrong answer

The bottleneck is Qiskit's own ordering strategy

What we checked	Reading Qiskit's current Rust source directly, not assuming
What we found	VF2Layout uses a custom search engine with a fixed, hardcoded node-ordering strategy — not the public rustworkx library many assume
Why it matters	That ordering processes symmetric, interchangeable circuit regions last — exactly the structure that saturates real devices
The fix that works	A single call to the public rustworkx library with the simplest possible ordering resolves every failing instance in a fraction of a millisecond

26 of 26 — verified, not just claimed

26 / 26

Layouts found

up from 0/26 before a defect fix we found and corrected

0

Coupling violations

every output checked directly against the device's own connectivity

0.99999999999999994

Minimum fidelity

36 independent checks — machine precision, not approximate

Verification also extends beyond the square-grid topology used in development — including one lattice (“brick”) on which the search's own earlier strategies had previously found nothing at all.

Gate synthesis: 7–9x faster, exactly correct

Independent of the layout work, PSF-Zero replaces Qiskit's general-purpose two-qubit gate synthesis with an analytic Cartan (KAK) decomposition, compiled to native Rust code.

- 7.0–9.3x cumulative speedup over Qiskit's default synthesis path, measured across three independent 10,000-iteration sessions
- Exact correctness: fidelity of 1.000000000000 at verifiable scale
- We also report what we could not explain — an unexplained variance change across sessions, investigated with the actual source diff and left honestly unresolved

CUMULATIVE SPEEDUP RANGE

7.0–9.3x

over Qiskit optimization_level=3,
same circuit family, three sessions

We report the honest range across sessions, not a single cherry-picked number — and disclose one figure we corrected ourselves.

Every claim was pre-registered and audited

Pre-registered

Predictions written and locked before every measurement — including ones later falsified

Independently repeated

Every headline figure reproduced across separate, independent runs

Self-audited

We found and corrected two of our own earlier claims — published alongside the originals, not hidden

Fully public

Raw data, scripts, and the complete chronological record, including corrections, are all on GitHub

The full technical record is public

PAPER 1

Ordering Sensitivity in Subgraph-Isomorphism Layout Search

Characterizes Qiskit's own failure region, identifies the mechanism, and reports what we could not yet explain.

arXiv preprint (in preparation)

PAPER 2

PSF-Zero: Gate Synthesis + a Verified, Ordering-Aware Layout Search

Documents the system built on Paper 1's finding, and its end-to-end verification.

arXiv preprint (in preparation)

Try it on your own circuits

We can run a short, no-commitment technical evaluation against a representative circuit — under NDA if the circuit itself is sensitive.

- 1 Share a representative circuit (or a sanitized equivalent)
- 2 We run PSF-Zero against it and report results directly
- 3 You decide whether it's worth a deeper evaluation

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